Individual Assignment: Machine Learning Tutorial

IMBIA MBOMENE NJEO

24149699

Table of Contents

Introduction 3
What Random Forest Is 3
Why the Number of Trees Is a Good Focus 4
Dataset Choice 4
Code Example 4
Interpreting the Results 7
Strengths and Limitations 8
Conclusion 9
References 10

Introduction

Random Forest is one of the most understandable models for teaching ensemble learning, combining a simple concept with strong practical results. It constructs large numbers of trees across multiple data samples and averages their predictions, thereby displacing the need for a single decision tree. The last answer, in classification, is the result of voting across trees; in regression, it is the result of averaging. The deeper explanation of why it works can be found in the original paper by Breiman: the overall error is determined by how powerful the individual trees are and how they are correlated with each other. Reduced correlation and moderately high trees tend to improve ensemble performance. The approach followed in this tutorial is a focused one, since nestimators are examined in detail as the number of trees in the forest increases. It is a powerful teaching theme since it presents a literal graphical account. As the number of trees grows, performance usually rises initially and then levels off, which helps learners understand not only the advantage but also the threshold usefulness of growing the models. The same argument is made by the out-of-bag example in scikit-learn: the error can be plotted against the number of trees, and the curve can be used to detect when performance becomes stable.

What Random Forest Is

A Random Forest is a meta-estimator, which is constructed by a large number of decision trees. In scikit-learn, the data is split into small blocks, and each tree is trained on bootstrap samples of these blocks, usually with stratified sampling. The tree predictions are averaged across a collection of trees to improve prediction and reduce overfitting (Georganos *et al.*, 2021). The trees themselves employ the standard best-split strategy, but because of their repetitive sampling and feature selection, the forest adds a dose of randomness. Breiman defined random forests as a set of tree predictors, in which each tree relies on a random vector sampled from a distribution identical to the one used by all trees. He further contended that, since the number of trees is kept large, the generalization error does not worsen indefinitely as the number of trees increases. That is why Random Forest is a beneficial alternative to a single decision tree, which could overfit the training data and be unstable. The model can be presented in terms of teaching, as it means relying on numerous professionals rather than depending wholly on one (Iranzad and Liu, 2025).

Why the Number of Trees Is a Good Focus

The parameter of estimators is the number of trees in the ensemble. The default that is recorded in scikit-learn is 100 trees. Even a small forest can, to some extent, act like a single wobbly tree, since the models are too few to even out the noise and instability. As forest size increases, the prediction tends to stabilize (Attanasi and Coburn, 2023). Nevertheless, there is a point beyond which gains decrease, and therefore, the additional computations required to achieve minor gains represent the dominant trade-off. This is what makes nestimators the best choice for a tutorial, since it yields a clean experiment. A learner can keep most parameters fixed, change only the number of trees, and observe changes in test accuracy. The curve, in most instances, increases rapidly at the beginning and then levels off. That trend reinforces Breiman's theoretical findings and the practical insights from scikit-learn's out-of-bag example: the initial trees are helpful, but at some point, the forest reaches diminishing returns.

Dataset Choice

The Wine data is a good choice for this tutorial because it is a small, well-structured dataset and is already built into scikit-learn. It is a three-classification data set consisting of 178 samples and 13 quantitative variables. The UCI Machine Learning Repository provides this data. It includes the outcomes of three chemical analyses of wines produced in the same region (Italy) but from three distinct cultivars (Salman, Kalakech and Steiti, 2024). The UCI also says it is a good network and well classified, making it a good dataset for testing a new classifier and, in turn, an ideal dataset for testing.

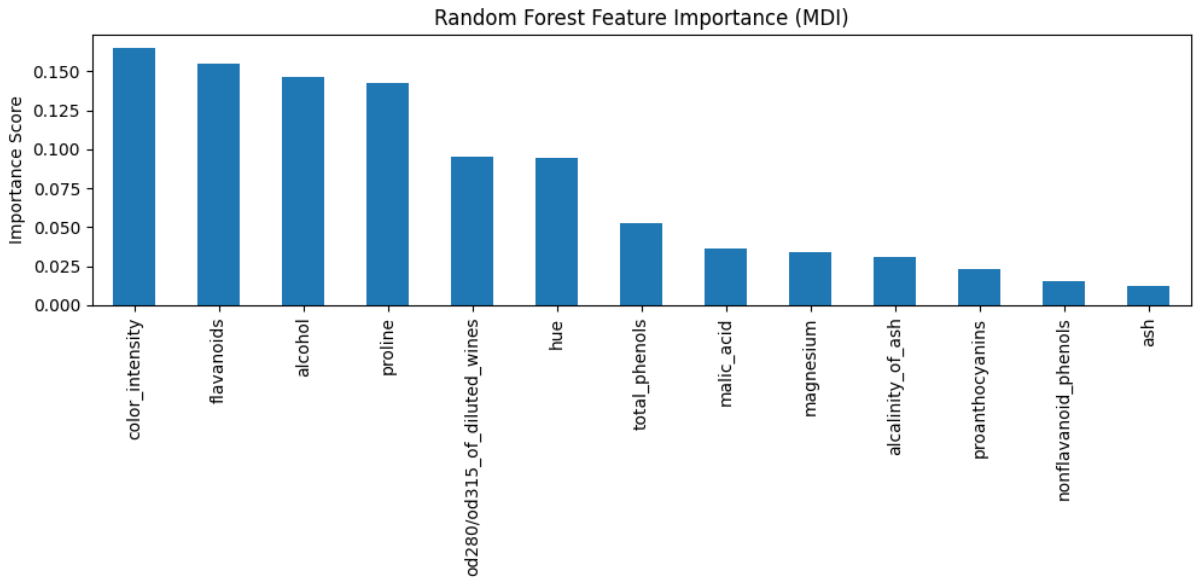
Code Example

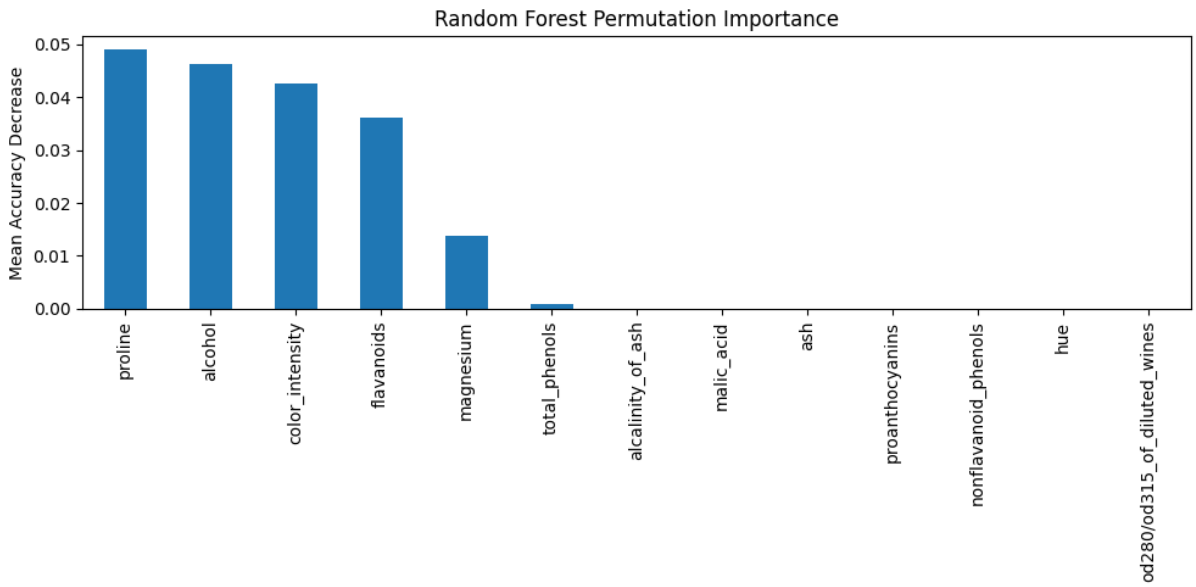
An example below utilizes the Wine dataset, as it is small and a multiclass dataset with named features, which makes it ideal for achieving both ensemble output and model interpretability. The code compares a single decision tree to a random forest, and analyses feature importance using both impurity-based and permutation-based methods. In scikit-learn, a Random Forest is an ensemble of decision trees trained on subsampled data, with predictions averaged or majority-voted to enhance accuracy and reduce overfitting (Avcı *et al.*, 2023). It is then followed by permutation of importance, which is a more inspection-oriented procedure that quantifies the reduction in model performance resulting from shuffling a feature.

```
import pandas as pd
import matplotlib.pyplot as plt
```

```python
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.inspection import permutation_importance
from sklearn.metrics import accuracy_score, classification_report

# Load dataset
wine = load_wine()
X = pd.DataFrame(wine.data, columns=wine.feature_names)
y = wine.target

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, random_state=42, stratify=y
)

# Single decision tree baseline
tree_model = DecisionTreeClassifier(random_state=42)
tree_model.fit(X_train, y_train)
tree_pred = tree_model.predict(X_test)
tree_acc = accuracy_score(y_test, tree_pred)

# Random Forest model
rf_model = RandomForestClassifier(
    n_estimators=200,
    max_depth=None,
    random_state=42
)
rf_model.fit(X_train, y_train)
rf_pred = rf_model.predict(X_test)
rf_acc = accuracy_score(y_test, rf_pred)

print("Decision Tree Accuracy:", round(tree_acc, 4))
print("Random Forest Accuracy:", round(rf_acc, 4))
print("\nRandom Forest Classification Report:\n")
print(classification_report(y_test, rf_pred,
target_names=wine.target_names))

# Impurity-based feature importance
mdi_importance = pd.Series(
    rf_model.feature_importances_,
    index=X.columns
).sort_values(ascending=False)

print("\nImpurity-Based Feature Importance (MDI):\n")
print(mdi_importance)

# Permutation importance
perm_result = permutation_importance(
    rf_model, X_test, y_test,
    n_repeats=20,
    random_state=42,
    scoring="accuracy"
)
```

```
perm_importance = pd.Series(
    perm_result.importances_mean,
    index=X.columns
).sort_values(ascending=False)

print("\nPermutation Feature Importance:\n")
print(perm_importance)

# Plot impurity-based feature importance
plt.figure(figsize=(10, 5))
mdi_importance.plot(kind="bar")
plt.title("Random Forest Feature Importance (MDI)")
plt.ylabel("Importance Score")
plt.tight_layout()
plt.show()

# Plot permutation importance
plt.figure(figsize=(10, 5))
perm_importance.plot(kind="bar")
plt.title("Random Forest Permutation Importance")
plt.ylabel("Mean Accuracy Decrease")
plt.tight_layout()
plt.show()
```





Interpreting the Results

As the results indicate, the Random Forest is more effective than the single decision tree on the test set. This result aligns with the principle of ensemble learning. One decision tree can be very close to the training data and highly sensitive to minor changes in the sample. However, a Random Forest minimises such instability by averaging trees trained on bootstrap groups and random feature subsets (Rhodes, Cutler and Moon, 2023). The model has a very clear description as a meta-estimator that fits a series of decision tree classifiers on sub-sets of the data and averages the results to enhance predictive accuracy and curb overfitting in scikit-learn. This interpretation aligns with Breiman's original work, in which he describes a forest as powerful when trees are not excessively correlated with one another and individual trees are useful but not too closely related. The second half of the tutorial, interpretability, is the output of the feature importance. The importance scores based on impurity, also called mean decrease in impurity, reflect the features that were most effective at subdividing the data across the forest. When any of the variables in the wine chemistry differ more distinctly from the other variables, it can be inferred that the model is more dependent on that measurement in separating the three wine classes (Fatima *et al.*, 2023). This interpretability is appealing in the teaching process, as it provides a brief preview of what the ensemble found to be influential. The example of forest importance provided by scikit-learn also shows that these values vary across trees, to the point that explainability in Random Forests does not hinge on a single tree but on trends across the ensemble.

However, the interpretation that is usually given by the permutation-importance plot is more conservative and practically significant. The principle of permutation importance is that each feature is randomised, and the change in the model score is measured. If accuracy drops drastically when a feature is shuffled, the model is obviously relying on that feature. This approach is particularly effective when dealing with non-linear models and opaque estimators, since it relates importance to predictive performance rather than to the internal dynamics of tree splitting (Nachouki *et al.*, 2023). The two plots can thus be used to serve different interpretive aims: impurity-based importance shows how often and how strongly features assisted in organising the forest, and permutation importance shows the extent to which the eventual predictor system relies on each feature.

Strengths and Limitations

Random Forest is an ideal teaching model with several strengths. It shows the reasoning behind organisational learning in a simple, articulate manner: several feeble or averagely powerful decision-makers can be more dependable in their overall decision-making than a model functioning in isolation. It can also be applied to tabular data, is flexible, and, in most cases, does not require the sensitive tuning required by several more sophisticated algorithms. The most important aspect of this task is that it provides access to interpretability through feature-importance analysis (Hu and Szymczak, 2023). This implies that it is not an instrument used exclusively to make predictions but also to deliberate on the variables important to the learning process. Breiman also maintained that random forests could not only give correct predictions but also provide useful information about the inner workings of the black box. Although these are the model's strengths, it has definite weaknesses. The importance based on impurity can be misleading, as it can overestimate the importance of numbers or high-cardinality features. Scikit-learn particularly warns about this and advises examining permutation importance as a more interpretable and reliable approach. Even the importance of permutation is limited because correlated features may mask one another. With multicollinear data, the model can achieve high accuracy, but permutation scores would be unacceptably small if several relevant variables are omitted due to feature overlap. Another weakness is that Random Forest is more interpretable than other complex models at a global level, but less interpretable at the individual decision level. It is possible to see which features are important at large, but the ultimate forecast appears to emerge from a myriad of trees rather than a single codified regulation (Nachouki *et al.*, 2023).

Conclusion

Overall, Random Forest offers a robust GIS that better explains ensemble learning and provides useful interpretability. The value of the ensemble is easily observed compared to a single decision tree, because most of the time the forest provides more stable, more accurate predictions by combining many less-correlated trees. The feature-importance analysis is then used to continue the lesson by revealing that it is not enough to look only at performance; it is also necessary to question which variables the model is utilising and the reliability of its interpretations. This kind of combination is particularly useful in teaching, as it exposes learners to both predictive thinking and model inspection. An effective tutorial thus not only introduces the concept of Random Forest as an efficient algorithm, but also demonstrates that it is a tool to bridge the gap between the intuitive nature of machine learning concepts and the responsibility of interpreting their findings.

References

Attanasi, E.D. and Coburn, T.C. (2023) 'Random Forest', in B.S. Daya Sagar, Q. Cheng, J. McKinley, and F. Agterberg (eds) *Encyclopedia of Mathematical Geosciences*. Cham: Springer International Publishing (Encyclopedia of Earth Sciences Series), pp. 1182–1185. Available at: https://doi.org/10.1007/978-3-030-85040-1_265.

Avcı, C., Budak, M., Yağmur, N. and Balçık, F. (2023) 'Comparison between random forest and support vector machine algorithms for LULC classification', *International Journal of Engineering and Geosciences*, 8(1), pp. 1–10.

Fatima, S., Hussain, A., Amir, S.B., Awan, M.G.Z., Ahmed, S.H. and Aslam, S.M.H. (2023) 'Xgboost and random forest algorithms: an in depth analysis', *Pakistan journal of scientific research*, 3(1), pp. 26–31.

Georganos, S., Grippa, T., Niang Gadiaga, A., Linard, C., Lennert, M., Vanhuysse, S., Mboga, N., Wolff, E. and Kalogirou, S. (2021) 'Geographical random forests: a spatial extension of the random forest algorithm to address spatial heterogeneity in remote sensing and population modelling', *Geocarto International*, 36(2), pp. 121–136. Available at: https://doi.org/10.1080/10106049.2019.1595177.

Hu, J. and Szymczak, S. (2023) 'A review on longitudinal data analysis with random forest', *Briefings in bioinformatics*, 24(2), p. bbad002.

Iranzad, R. and Liu, X. (2025) 'A review of random forest-based feature selection methods for data science education and applications', *International Journal of Data Science and Analytics*, 20(2), pp. 197–211. Available at: https://doi.org/10.1007/s41060-024-00509-w.

Nachouki, M., Mohamed, E.A., Mehdi, R. and Abou Naaj, M. (2023) 'Student course grade prediction using the random forest algorithm: Analysis of predictors' importance', *Trends in Neuroscience and Education*, 33, p. 100214.

Rhodes, J.S., Cutler, A. and Moon, K.R. (2023) 'Geometry-and accuracy-preserving random forest proximities', *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(9), pp. 10947–10959.

Salman, H.A., Kalakech, A. and Steiti, A. (2024) 'Random forest algorithm overview', *Babylonian Journal of Machine Learning*, 2024, pp. 69–79.

Sun, Z., Wang, G., Li, P., Wang, H., Zhang, M. and Liang, X. (2024) 'An improved random forest based on the classification accuracy and correlation measurement of decision trees', *Expert Systems with Applications*, 237, p. 121549.